

QLORA: Efficient Finetuning of Quantized LLMs

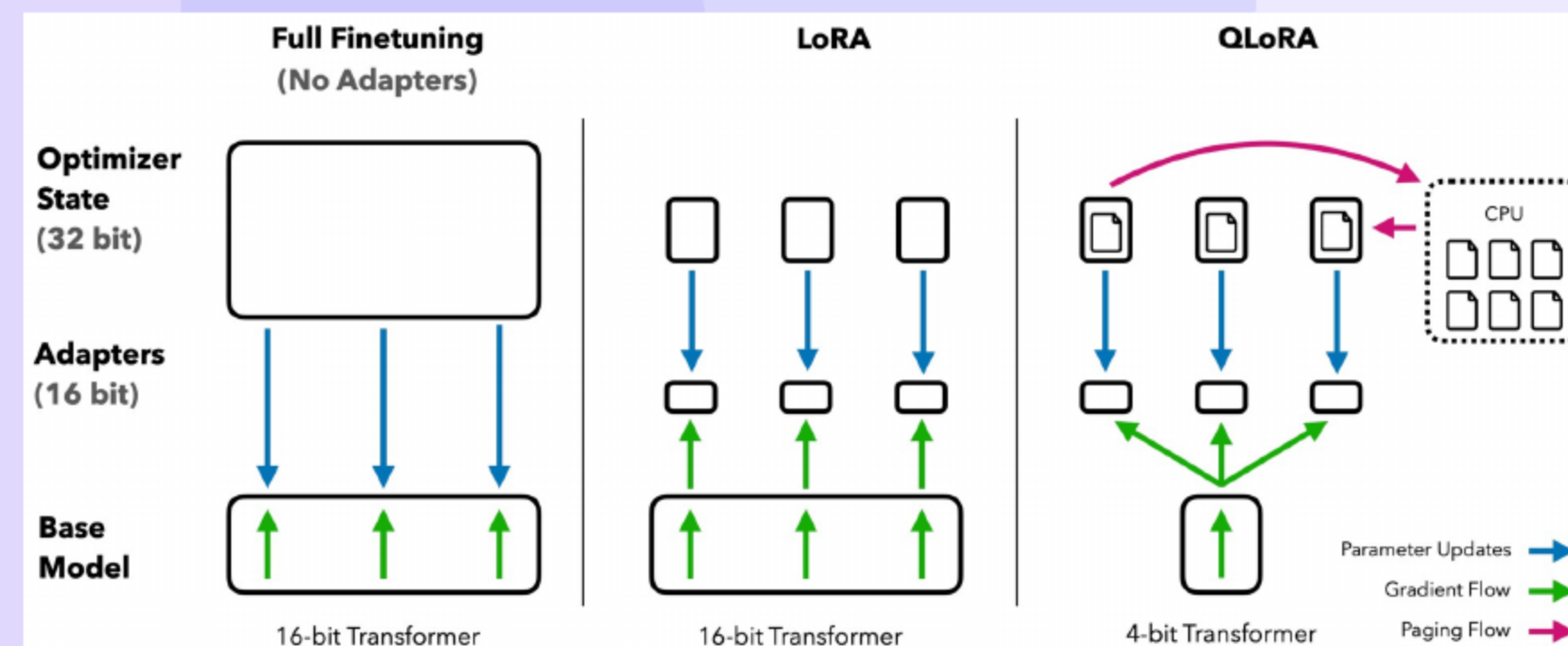
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01 QLORA: Title and Authors

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- QLORA: Efficient finetuning of quantized large language models

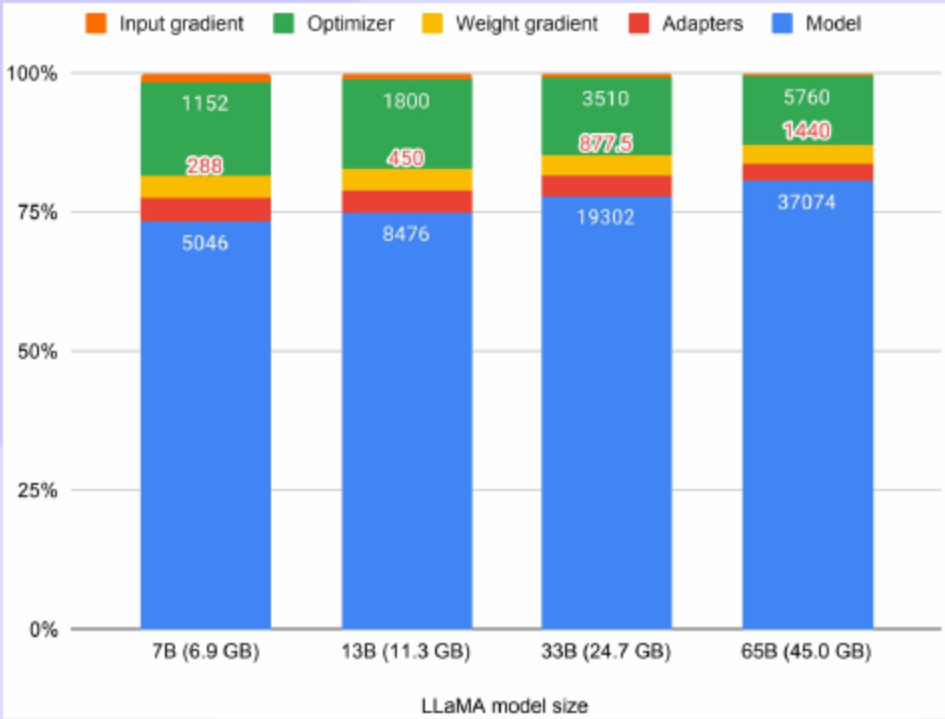
02 QLORA: What, How, Outcomes



- QLORA backpropagates through a frozen 4-bit base into LoRA adapters with full 16-bit parity
- NF4 quantization, double quantization, and paged optimizers enable stable finetuning
- Guanaco achieves [near-ChatGPT quality](#) and scales to 65B on one 48 GB GPU

03 Memory Problem and Approach

- Sixteen-bit finetuning of 65B needs more than 780 GB and risks **out-of-memory spikes**
- QLORA reduces working memory to under 48 GB using 4-bit NF4 and paged optimizers
- Training updates apply only to LoRA while compute runs in BF16 after dequantization



- 24 GB: 7B/13B; 48 GB: 65B with QLORA
- Memory driven by params, optimizer states, activations
- Grad accumulation and paged optimizers prevent spikes

- QLORA cuts memory from more than 780 GB to under 48 GB with parity to 16-bit finetuning
- Guanaco 65B reaches [near-ChatGPT performance](#) within roughly one day
- NF4, double quantization, and paged optimizers drive **efficiency** and **stability**

- Sixteen-bit finetuning is memory prohibitive for 65B and suffers **peak usage spikes**
- Naive low-bit training destabilizes gradients and leaves memory as the bottleneck
- QLORA freezes 4-bit weights and trains LoRA with **stable** and **efficient** updates

- Inference quantization methods improve forward fidelity but rarely address training stability
- Backpropagation through quantized weights is explored at scale and aligns with QLORA design
- Parameter-efficient finetuning options exist, while LoRA offers **robust** and **simple** integration

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Why Prior Quantization Fails

- Block-wise quantization improves bin use but leaves gradient flow unstable during training
- Outliers and coarse bins distort updates and increase **training variance**
- Training-aware design is required to preserve **stability** and **accuracy**

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Adapters Don't Solve Memory

- LoRA parameter counts are small while activations and gradients dominate memory
- Gradient checkpointing is essential and enables multiple adapters with low overhead
- Reducing adapter size yields limited savings compared to **activation memory**

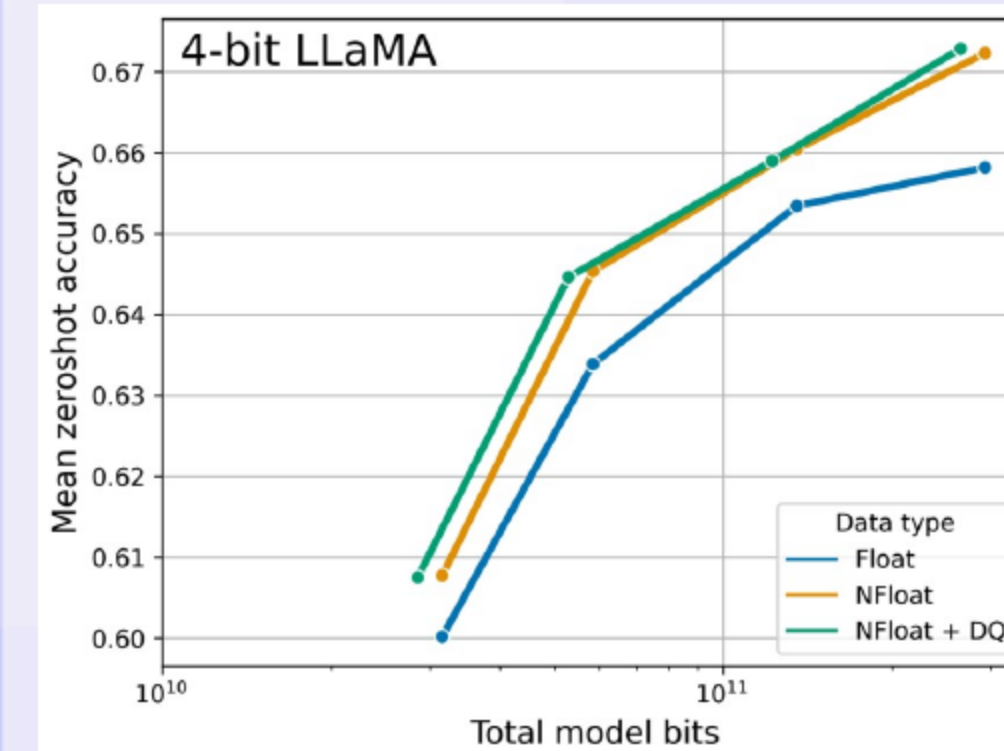
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Study Scope and Themes

- More than one thousand finetunes span 80M to 65B across models and datasets
- QLORA matches strong 16-bit baselines and powers Guanaco chatbots
- Data quality outweighs size for **instruction following** performance

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NF4 Quantization



- NF4 uses fixed quantiles for zero-centered weights and provides an exact zero level
- Weights are stored in 4-bit and dequantized to BF16 for computation
- NF4 outperforms other 4-bit types and is robust to outliers

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DQ + Paged Optimizers

- Double quantization compresses scales, reducing bits per parameter
- Paged optimizers use unified memory to avoid OOM
- Lower overhead preserves quality, enables larger models

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Training Recipe

- Weights are stored in NF4 with dequantization of scales and computation in BF16
- Backpropagation updates only LoRA parameters while base weights remain frozen
- One storage type and one compute type simplify **engineering** and **deployment**

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Guanaco and MMLU

- Guanaco 65B: near-ChatGPT on Vicuna; strong Elo
- FLAN v2 improves MMLU, aligns poorly with chat metrics
- Small high-quality data beats larger mixed sets

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NF4 Evidence and Parity

- NF4 beats FP4/Int4 on accuracy and perplexity.
- Four-bit QLORA matches strong 16-bit baselines.
- Memory efficiency lets larger 4-bit models win.

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Comparison: **16-bit** Baseline

- GLUE: QLORA matches 16-bit (≈ 88.6 – 88.8)
- MMLU: NF4 $\approx$ 16-bit; FP4 ~ 1 point behind
- Memory: 780 GB vs <48 GB (65B finetuning)
- Guanaco 65B trains ≈ 24 h on one GPU

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Protocols and Setups

- Benchmarks cover GLUE, TKInstruct, and LLaMA from 7B to 65B for generative tasks
- Training uses NF4 with double quantization and paged optimizers across all linear layers
- Evaluation includes Vicuna and Open Assistant with human and GPT-4 pairwise judgments

[GLUE]; [TKInstruct]; [Vicuna]; [OpenAssistant]; [GPT-4]

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Ablations That Matter

- Placing adapters in all linear layers best preserves parity with full finetuning
- LoRA rank between eight and thirty-two shows similar quality while very small ranks underfit
- NF4 block size sixty-four balances **accuracy** and **overhead** effectively

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Qualitative Biases and Limits

- Models show strong recall on common facts and **overconfidence** on obscure items
- Automated judges exhibit order bias and **self-preference** in pairwise scoring
- Human agreement is moderate and varies with system strength and task type

- Verification at 33B and 65B remains **resource intensive** across broader tasks
- Coverage omits several benchmarks and limits generalization claims
- Unexplored precisions and adapters suggest **future headroom** for efficiency

[BigBench]; [RAFT]; [HELM]

- Democratized finetuning improves **access** and supports transparent research
- Lower barriers can **amplify misuse** without safeguards and oversight
- Mitigations include model cards, provenance, filters, and targeted red-teaming

[Model Cards]

- QLORA enables 33B on consumer hardware and 65B on a single 48 GB GPU with parity
- Released models and code support **rapid adoption** and reproducibility
- On-device finetuning of 7B-class models becomes feasible for private applications

[Code Release]

Thank You

Questions & Discussion